

Stats 210A, Fall 2026

Homework 10

Due on: Wednesday, Nov. 18

Instructions: See the standing homework instructions on the course web page

Problem 1 (James-Stein estimator with regression-based shrinkage). Consider estimating $\theta \in \mathbb{R}^n$ in the model $Y \sim N_n(\theta, \sigma^2 I_n)$. In the standard James-Stein estimator, we shrink all the estimates toward zero, but it might make more sense to shrink them towards the average value $\bar{Y}$ (as we explored in a previous problem) or towards some other value based on observed side information.

Suppose that we have side information about each parameter θ_i , represented by covariate vectors $x_1, \dots, x_n \in \mathbb{R}^d$. Assume the design matrix $X \in \mathbb{R}^{n \times d}$, whose i th row is x'_i , has full column rank. Suppose that we expect θ_i is not too far from $x'_i \beta$ for some $\beta \in \mathbb{R}^d$. But unlike the usual linear regression setup, we will not assume $\theta_i = x'_i \beta$ exactly, we just want to shrink our estimate toward $x'_i \beta$.

- (a) Assume the error variance $\sigma^2 = 1$ is known. Find an estimator $\delta(Y)$ for θ that strictly dominates $\delta_0(Y) = Y$ whenever $n - d \geq 3$,

$$\text{MSE}(\theta; \delta) < \text{MSE}(\theta; \delta_0), \quad \text{for all } \theta \in \mathbb{R}^n,$$

and for which $\text{MSE}(X\beta; \delta) = d + 2$, for any $\beta \in \mathbb{R}^d$.

In the special case of “intercept-only” regression ($d = 1$ and $x_i = 1$ for all i), your estimator should reduce to the version of the James-Stein estimator that shrinks toward $\bar{Y}$ (but you do not have to show this).

Hint: The problem will become easier after an appropriate change of basis; think about how the estimator operates on different subspaces.

- (b) Continue to assume the error variance $\sigma^2 = 1$ is known. Suppose we are unsure of whether $\theta = X\beta$ exactly. Suggest an appropriate test of the hypothesis $H_0 : \theta = X\beta$ vs $H_1 : \theta \neq X\beta$, treating $\beta \in \mathbb{R}^d$ as an unknown nuisance parameter.
- (c) **Optional:** (Not graded, no extra points) Now suppose that the error variance $\sigma^2 > 0$ is unknown, but we have $r > 1$ replicates for each i ; that is, we observe $Y_{i,k} \stackrel{\text{i.i.d.}}{\sim} N(\theta_i, \sigma^2)$ for $i = 1, \dots, n$ and $k = 1, \dots, r$. Modify your test from the previous part for $H_0 : \theta = X\beta$ vs $H_1 : \theta \neq X\beta$.

Moral: We can get interesting estimators by combining our change of basis ideas with the James–Stein estimator, to implement different sorts of inductive biases.

Problem 2 (Confidence regions for regression). Assume we observe $x_1, \dots, x_n \in \mathbb{R}$, which are not all identical (for at least one pair i and j , $x_i \neq x_j$). We also observe

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \text{for } \varepsilon_i \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

$\beta_0, \beta_1 \in \mathbb{R}$ and $\sigma^2 > 0$ are unknown. Let $\bar{x}$ represent the mean value $\frac{1}{n} \sum_i x_i$.

- (a) Give an explicit expression for the t -based confidence interval for β_1 , in terms of a quantile of a Student’s t distribution with an appropriate number of degrees of freedom (feel free to break up the expression, for example by first giving an expression for $\hat{\beta}_1$ and then using $\hat{\beta}_1$ in your final expression).

- (b) Define the OLS estimator $\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix}$. Show that $\hat{\beta} \sim N_2(\beta, \sigma^2(X'X)^{-1})$, for the design matrix $X = [1_n, x]$. Apply this fact to find an F -test for the hypothesis $H_0 : \beta = 0$ vs $H_1 : \beta \neq 0$.
- (c) Invert your F -test to give a *confidence ellipse* for $\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}$. It may be convenient to represent the set as an affine transformation of the unit ball in $\mathbb{R}^2$:

$$b + A\mathbb{B}_1(0) = \{b + Az : z \in \mathbb{R}^2, \|z\| \leq 1\}, \quad \text{for } b \in \mathbb{R}^2, A \in \mathbb{R}^{2 \times 2}.$$

Give explicit expressions for b and A in terms of a quantile of an appropriate F distribution.

Moral: Inverting t - and F -tests is a very general tool for constructing confidence regions. Similar ideas will come up in future lectures for multivariate Gaussian random estimators.

Problem 3 (Confidence bands for regression). The setup for this problem is the same as for the previous problem only now we are interested in giving *confidence bands* for the regression line $f(x) = \beta_0 + \beta_1 x$. In this problem you do not need to give explicit expressions for everything, but you should be explicit enough that someone could calculate the bands based on your description.

- (a) For a fixed value $x_0 \in \mathbb{R}$ (not necessarily one of the observed x_i values) give a $1 - \alpha$ t -based confidence interval for $f(x_0) = \beta_0 + \beta_1 x_0$. That is, we want to find $C_1^P(x_0), C_2^P(x_0)$ such that

$$\mathbb{P}(C_1^P(x_0) \leq f(x_0) \leq C_2^P(x_0)) = 1 - \alpha.$$

For each x_0 , the coverage should be exactly $1 - \alpha$. The functions $C_1^P(x), C_2^P(x)$ that we get from performing this operation on all x values give a *pointwise confidence band* for the function $f(x)$.

- (b) Now give a *simultaneous confidence band* around $f(x) = \beta_0 + \beta_1 x$. That is, give $C_1^S(x), C_2^S(x)$ with

$$\mathbb{P}(C_1^S(x) \leq f(x) \leq C_2^S(x), \text{ for all } x \in \mathbb{R}) \geq 1 - \alpha,$$

and show that your confidence band has this property.

Hint: If all we know is that β is in the confidence ellipse from the previous problem, what can we deduce about $f(x)$?

- (c) Download the data set in `hw10.csv` from the course web site and make a scatter plot of the data. Plot the OLS regression line as well as the two confidence bands. Describe what you see. What do the bands do as x goes away from the data set, and why does this make sense?
- (d) **Optional:** (Not graded, no extra points) Show that the coverage of the simultaneous confidence band is *exactly* $1 - \alpha$, not just greater than or equal to $1 - \alpha$.

Moral: This is an example of a “deduced inference” whereby we arrive at simultaneous confidence regions for many (even infinitely many in this case) estimands at once, by assuming that an initial confidence region covers and then “deducing” all possible conclusions that follow from that assumption. So long as the initial confidence region covers, our deductions will all be simultaneously correct.

Problem 4 (Probabilistic big-O notation). Let $X_1, X_2, \dots$ denote a sequence of random vectors (with $\|X_n\| < \infty$ almost surely for each n). We say the sequence is *bounded in probability* (or sometimes *tight*) if for every $\varepsilon > 0$ there exists a constant $M_\varepsilon > 0$ for which

$$\mathbb{P}(\|X_n\| > M_\varepsilon) < \varepsilon, \quad \forall n.$$

Informally, there is “no mass escaping to infinity” as n grows. Like regular big-O notation, these symbols can help to make rigorous asymptotic proofs look clean and intuitive.

For a fixed sequence a_n , we say $X_n = o_p(a_n)$ if $X_n/a_n \xrightarrow{P} 0$ as $n \rightarrow \infty$, and $X_n = O_p(a_n)$ if the sequence $(X_n/a_n)_{n \geq 1}$ is bounded in probability.

Prove the following facts for $X_n, Y_n \in \mathbb{R}^d$:

- (a) If $X_n \Rightarrow X$ for any random vector X , then $X_n = O_p(1)$.
- (b) If $X_n = o_p(a_n)$ then $X_n = O_p(a_n)$.
- (c) If $X_n = O_p(a_n)$ and $Y_n = o_p(b_n)$, then $X_n' Y_n = o_p(a_n b_n)$. If $X_n = O_p(a_n)$ and $Y_n = O_p(b_n)$, then $X_n' Y_n = O_p(a_n b_n)$.
- (d) If $X_n = O_p(1)$ and $g : \mathbb{R}^d \rightarrow \mathbb{R}^k$ is continuous then $g(X_n) = O_p(1)$.
- (e) For $d = 1$, if $X_n = O_p(a_n)$ with $a_n \rightarrow 0$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ is continuously differentiable with $g(0) = \dot{g}(0) = 0$, then $g(X_n) = o_p(a_n)$. Show further that if g is twice continuously differentiable then $g(X_n) = O_p(a_n^2)$. (**Hint:** Use the mean value theorem and apply a previous part of this problem.)
- (f) For $d = 1$, if $\text{Var}(X_n) = a_n^2 < \infty$ and $\mathbb{E}X_n = 0$ then $X_n = O_p(a_n)$. (**Hint:** Use Chebyshev's inequality.)
- (g) If $\text{Var}(X_n) = a_n^2 < \infty$, is it impossible to have $X_n = o_p(a_n)$? Prove or give a counterexample.

Moral: This notation is especially handy for dealing with error terms.